

NFAC Scoring — Calculation Insight

How every field input becomes a score. This document is the calculation knowledge base behind the analytics dashboard and the per-property PDF report: the exact formulas, deduction tiers, rubric weights, and a fully worked example from the evaluation dataset — so no scoring step is ever a black box.

The Model at a Glance

Six categories are each scored out of 100, then averaged. Every category is driven by concrete, observable field inputs captured in the on-site evaluation form.

Category	Raw scale	Weight to /100	Driven by
Safety & Risk	Severity tier	Direct (tier value)	Equipment inspection findings mapped to severity tiers
Asset Condition	5 metrics × (1–5) = /25	× 4	Operational readiness, cosmetics, lifecycle, maintenance, stewardship
User Experience	4 metrics × (1–5) = /20	× 5	Cleanliness, comfort/environment, layout, aesthetics
Utilization	4 metrics × (1–5) = /20	× 5	Wear patterns, equipment selection, space use, throughput
Exercise Coverage	10 modalities × (0/1) = /10	× 10	Presence of core movement categories
Sanitation Readiness	1 rubric (1–5) = /5	× 20	Sanitizer + wipe availability, stock, placement

FINAL NFAC SCORE = (Safety + Asset + Experience + Utilization + Coverage + Sanitation) ÷ 6

Grade bands: A ≥ 90 · B 80–89 · C 70–79 · D 60–69 · F < 60. Percentile rank is computed against every prior evaluation in the benchmark dataset, filtered by property type and market.

1 · Safety & Risk — Severity Tier Deduction

The score starts from a perfect state and is set by the worst finding class present. Every yes/no inspection answer (belt test, cable condition, e-stop, structure, power sharing, debris) maps to a tier. The category score is the tier value of the most severe condition found:

Tier	Score	Trigger condition (what the evaluator saw)
Excellent	100	No findings
Attention needed	94	Minor findings only (e.g., lubrication lacking, debris present)
Moderate risk	84	Major finding present (e.g., belt hesitates slightly under load)
Urgent risk	69	One critical finding (e.g., belt slows significantly; frayed cable, steel exposed)
Immediate risk	49	Multiple critical findings
Critical risk	25	Immediate life-safety hazard (e.g., compromised steel cable, belt stops under load)

The NFAC Safety Matrix in the dashboard is the visual form of the same inputs: each hazard family (high-speed equipment, cable systems, structural, electrical, fall hazards, user controls) shows Pass / Major / Critical based on the underlying checklist answers.

2 · Asset Condition — 5 Metrics × 4

Metric (1–5)	5 means	1 means
Operational Readiness	Every machine fully operational	Significant portion unusable / out of order
Cosmetic Condition	Like new	Poor appearance influencing perception
Equipment Lifecycle	Modern portfolio, little capital planning	End-of-life portfolio, large capex approaching
Preventative Maintenance	Consistent professional maintenance evident	Obvious neglect
Asset Stewardship	Proactive care of equipment and facility	Extensive deferred maintenance

Calculation: sum of the five metrics (max 25) × 4 → score out of 100.

3 · User Experience — 4 Metrics × 5

Cleanliness & Care · Comfort & Environment · Layout & Accessibility · Aesthetic Appeal — each rated 1–5 against written rubrics (e.g., Layout 5 = excellent traffic flow, ADA-conscious spacing; 1 = unsafe spacing). Sum (max 20) × 5 → /100.

4 · Utilization — 4 Metrics × 5

Wear Pattern Analysis · Equipment Selection Effectiveness · Space Utilization · Throughput Capacity — each 1–5. Throughput is judged against property size (e.g., an apartment at 200–350 units or a hotel at 100 rooms has a defined expected workout capacity). Sum (max 20) × 5 → /100.

5 · Exercise Coverage — 10 Modalities × 10

One point for each core movement category present: treadmill, elliptical, stationary bike, stepmill, rower, free weights, chest/multi press, leg curl/extension, lat/low pulldown, leg press (a functional trainer counts for the strength stations). Count (max 10) × 10 → /100.

6 · Sanitation Readiness — Rubric × 20

Single 1–5 rubric: 5 = both hand sanitizer and wipes readily available, stocked, well distributed; 3 = both present but placement/quantity lacking; 1 = neither present. Score × 20 → /100.

Worked Example — Broadstone Archive

(from the evaluation dataset)

Actual figures from the NFAC Scoring Framework v2 sample grade. Every number below traces directly to a field input recorded on site.

Category	Field inputs recorded	Raw	Weight	Score /100
Safety & Risk	Multiple critical findings: treadmill belts unlubricated (hesitate under load), non-responsive console → Immediate-risk tier	tier	direct	49
Asset Condition	Readiness 4 + Cosmetic 3 + Lifecycle 3 + Maintenance 3 + Stewardship 4	17 / 25	× 4	68
User Experience	Cleanliness 5 + Comfort 4 + Layout 4 + Aesthetics 5	18 / 20	× 5	90
Utilization	Wear 5 + Selection 5 + Space 4 + Throughput 5	19 / 20	× 5	95
Exercise Coverage	All 10 movement categories present	10 / 10	× 10	100
Sanitation Readiness	Sanitizer + wipes fully stocked and distributed	5 / 5	× 20	100

Final calculation	
Sum of category scores	49 + 68 + 90 + 95 + 100 + 100 = 502
Divide by 6	502 ÷ 6 = 83.67
Grade band	83.67 → B- (80–89 = B band, lower third)
Benchmark percentile	83.67 vs. all prior DFW apartment evaluations → 82nd percentile

Don't-Miss-A-Mark Checklist (calculation integrity)

These are the specific points the automation highlights on every evaluation so no scoring step is missed:

#	Checkpoint	Why it matters
1	Every equipment inspected has a severity tier assigned	One missed critical finding inflates Safety by up to 75 points
2	All 5 Asset Condition metrics scored (no blanks)	A blank metric silently drops the category by up to 20 points
3	All 4 UX and 4 Utilization metrics scored	Same blank-input risk; the engine blocks submission if missing
4	10/10 modality checklist completed	Coverage is the easiest category to under-count on site
5	Sanitation rubric selected (1–5), never inferred	×20 weight — a 1-point error moves the final score 3.3 points
6	Photo evidence attached for every non-Pass finding	Backs every deduction in the customer-facing report

#	Checkpoint	Why it matters
7	Property type + unit count recorded	Required for correct benchmark peer group and throughput rubric

Where These Numbers Surface

Analytics dashboard: executive score dial, six category donuts, score-composition bars (each 1–5 metric visualized), NFAC Safety Matrix, benchmark percentile bar, and portfolio comparisons. PDF report: the same figures rendered as a branded, print-ready document generated automatically from each form submission — dashboard and PDF always agree because both read the same calculation output.

Live demo: <https://userdevaccount1.github.io/premier-fitness-proposal/>